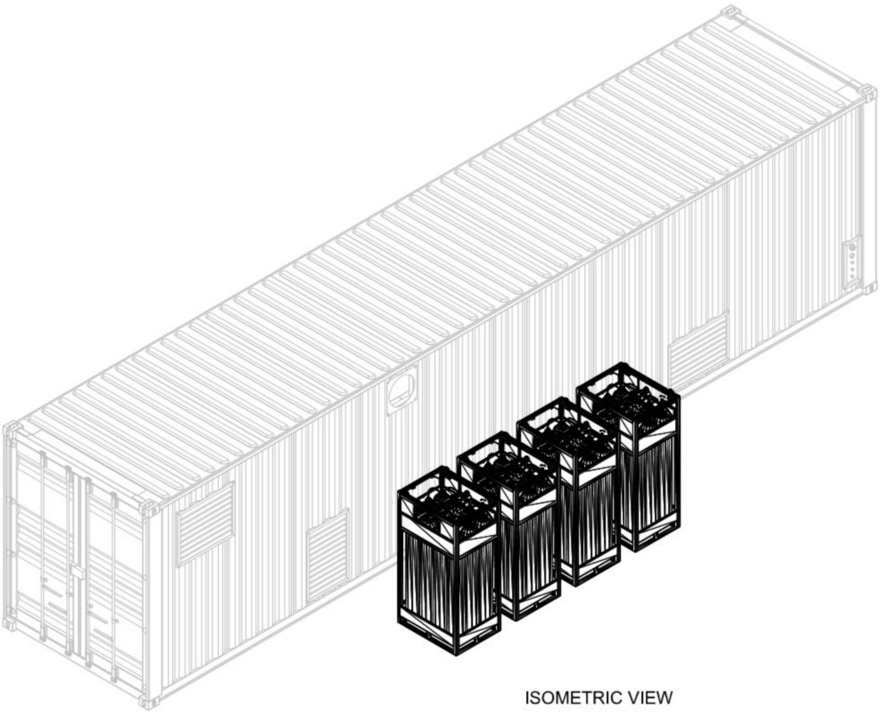
HORIZON
ENGINEERING
SOLUTIONS

Containerised Medical Grade Oxygen Generation Systems



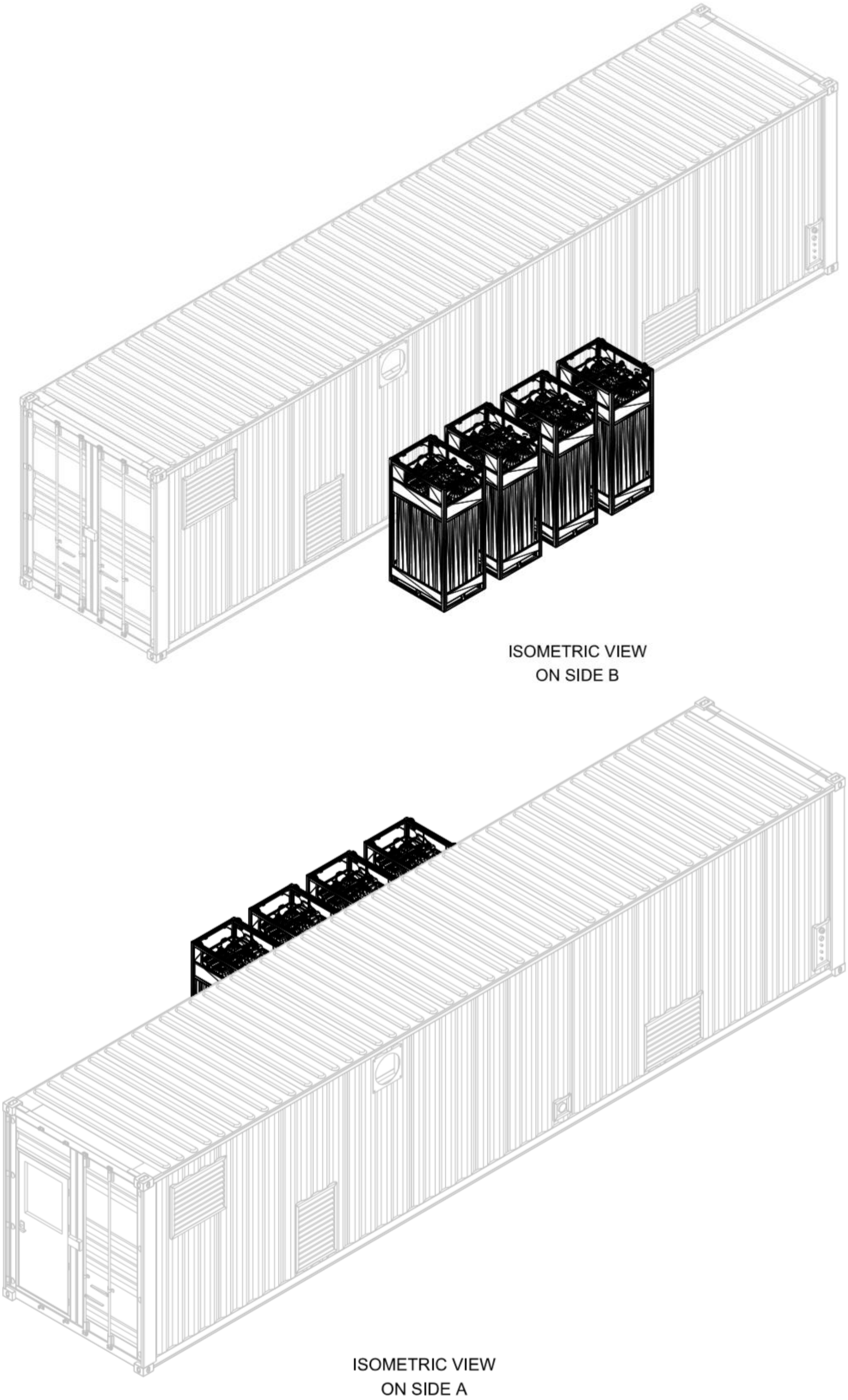
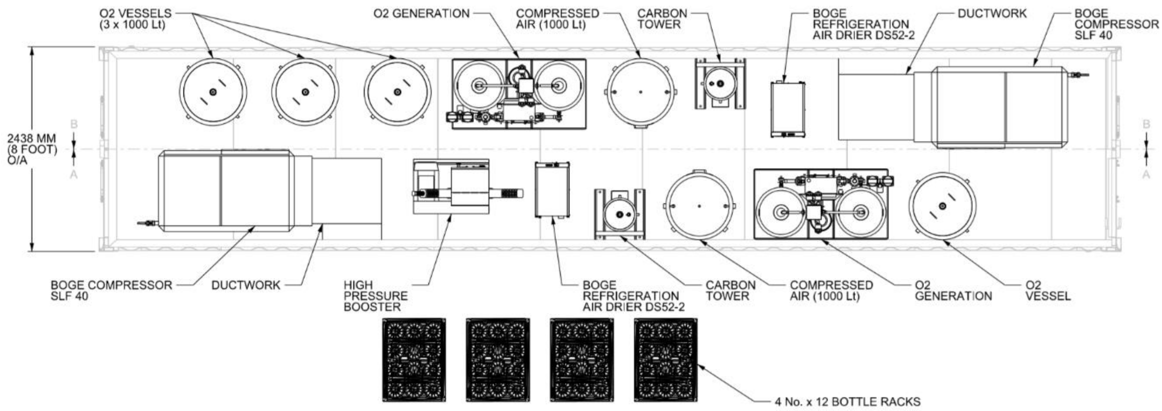
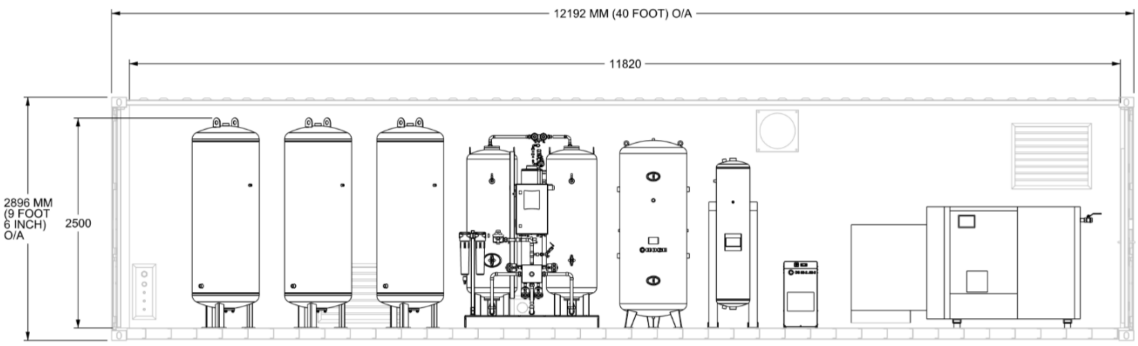


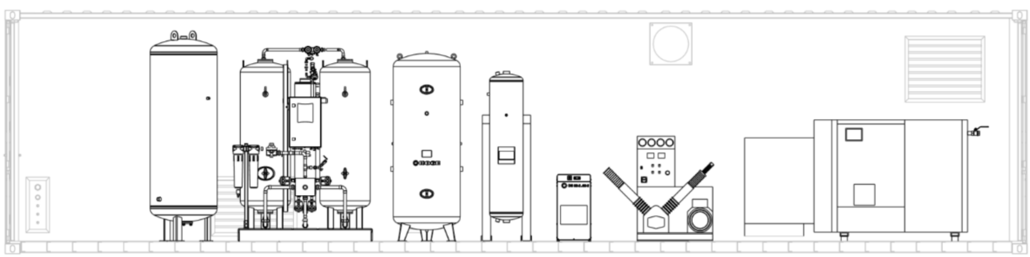
Figure 1: Isometric views of containerised PSA oxygen generation system



PLAN VIEW
(CONTAINER ROOF REMOVED FOR CLARITY)



SECTION VIEW A-A

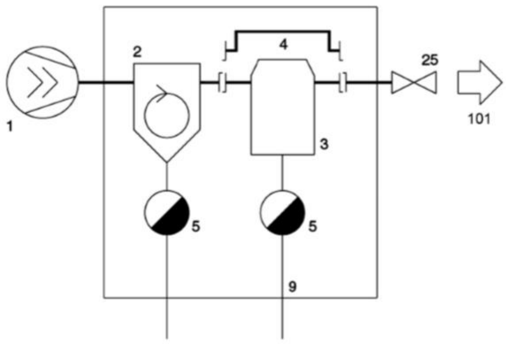


SECTION VIEW B-B

Figure 2: General layout drawings of containerised PSA oxygen generation system

Air Compressors:

The high pressure feed air for each PSA generator is provided by a Boge model SLF 40-3 oil lubricated compressor. Air is drawn via a louvre in the side wall of the container and is ducted to the compressor intake. This air is filtered in compliance with BS ISO 5011:2000 prior to entering the compressor. The air is then compressed to circa 7 Bar. Entrained compressor oil is removed from the compressed air by an internal oil separator and is returned to the compressor housing. Mechanical heat of compression is removed from the compressed air by an after cooler and is rejected externally from the container. This ensures that the temperature of the air entering the PSA generator is maintained at an acceptable level to prevent damage to the zeolite in the generator towers, and also reduces the build up of heat within the container.



101 Compressed air OUTLET

1 Screw compressor (see also flow diagram)
2 Cyclone separator
3 Refrigerant compressed air dryer
4 Bypass
5 Drain
9 Condensate
25 Non-return valve for compressed air exhaust

1 Intake filter
The intake filter cleans the air suctioned by the air end.

2 Intake regulator
The intake regulator opens (load operation) or closes (idling operation or standstill) the suction line depending on the operating condition of the compressor.

3 Air end
The air end compresses the air which is sucked in.

4 Compressed air-oil separator vessel
The compressed air separates from the oil under the force of gravity in the compressed air/oil vessel.

5 Oil separator
The oil separator separates the residual oil contained in the compressed air.

6 Minimum pressure non-return valve
The minimum pressure non-return valve does not open until the system pressure has increased to 3.5 bar. This causes a rapid build-up of the system pressure and ensures lubrication in the starting phase. Once the compressor has been switched off, the check valve prevents the compressed air from flowing back out of the mains line.

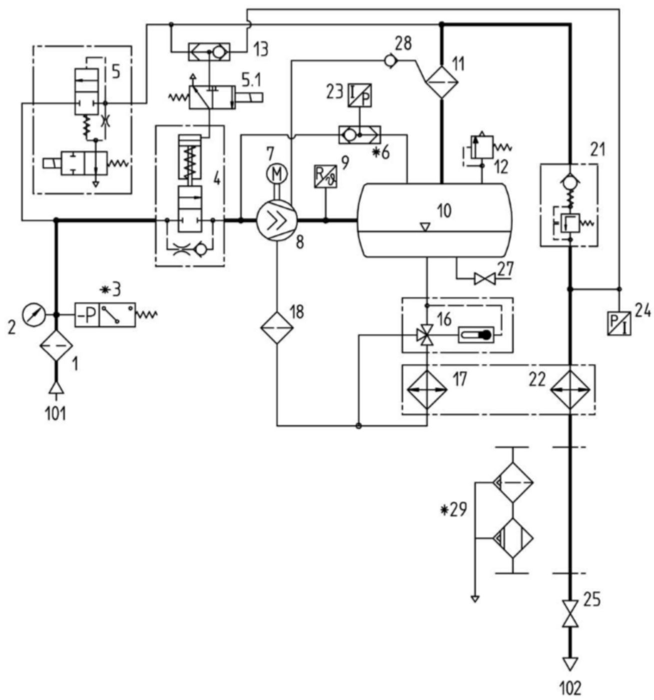
7 Compressed air aftercooler (air cooled or water cooled)
The compressed air is cooled in the compressed air aftercooler, causing the water contained in the air to condensate.

8 Stop valve
The screw compressor can be isolated from the mains using the stop valve.

Figure 3: Simplified description of in-feed air compression and drying system

The compressed air output of the compressor is regulated by varying the compressor motor speed using a variable speed drive, in response to hospital oxygen demand. The compressor is a fully packaged unit, with onboard controls and operator graphical display unit, which displays all compressor operating run and fault information and allows operational adjustments to be made.

The P&ID below details the configuration of the packaged air compressor. Full technical and maintenance details are provided in the manufacturer manuals in the handover documentation.



101 Intake air INLET
102 Compressed air OUTLET

1 Suction filter
2 Maintenance display
3 Suction filter monitoring *
4 Suction regulator
5 Unloading valve
5.1 Suction regulator control valve
6 Rotational direction monitoring with double non-return valve
7 Drive motor
8 Compressor air end
9 Sensor for final compression temperature
10 Combined compressed air/oil receiver
11 Oil separator
12 Safety valve
13 Quick start valve
16 Thermostatic oil control valve
17 Oil cooler
18 Oil filter
21 Minimum pressure non-return valve
22 Compressed air aftercooler
23 Pressure transmitter for system pressure
24 Pressure transmitter for network pressure
25 Stop valve for compressed air outlet
27 Oil drain
28 Non-return valve for drainage line
29 Cyclone separator with condensate drain and refrigerant compressed air dryer with condensate drain *

* Optional accessory

Figure 2: Detailed P&ID of Boge feed air compressors

Oxygen concentrators or pressure swing adsorber (PSA) systems are an alternative to the more traditional liquid or cryogenic oxygen supply systems (the terms oxygen concentrator and PSA are interchangeable). When installed, a PSA system will deliver product gas via the “oxygen” pipeline system.

Oxygen concentrators operate by adsorbing, under pressure, other gases in the atmosphere onto materials which have specific physico-chemical properties, thus freeing the oxygen which is stored and transmitting it for use. The adsorbents are known as artificial zeolites, more commonly referred to as molecular sieves. The sieve units are arranged in pairs, one adsorbing whilst the other regenerates. The waste product, essentially nitrogen, is discharged to atmosphere during regeneration of the adsorbents. Regeneration requires the use of a small proportion of the product gas.

The PSA process has reached a high level of technical sophistication and is capable of producing oxygen with a concentration of about 95%. The remainder is mainly argon with some nitrogen. The highest concentration is not likely to exceed 97/98%.

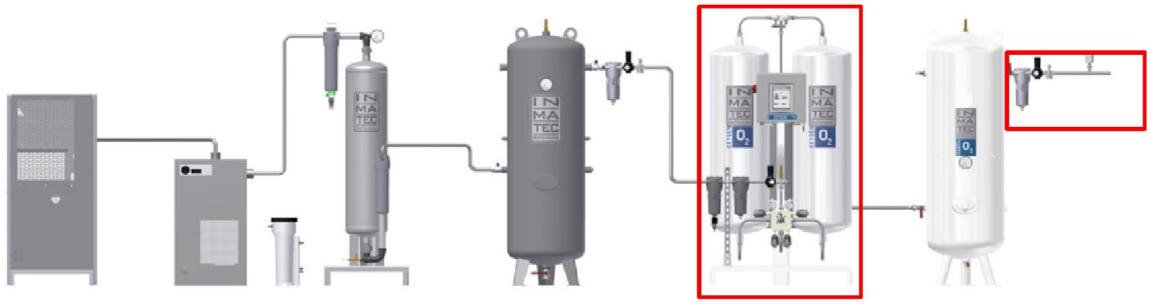


Fig. 1: Basic structure of an oxygen generation system

The major components of a PSA system and their layout are shown in Figure 1. These are typically: the compressor, refrigeration dryer, carbon tower/air filters, air receivermolecular sieves, oxygen filters and regulators.

INMATEC IMT POC med oxygen generators produce medical oxygen with an oxygen purity of 95% ± 3% from compressed air. For this purpose, the pressurised interchangeable tanks with a molecular sieve are fed with compressed air at 7 bar. Air compressors generate the compressed air required for this purpose.

Principle of Oxygen Generation:

INMATEC IMT POC med oxygen generators are based on pressure swing adsorption (PSA) technology. Pressure swing adsorption (PSA) technology uses a physical process for selective decomposition of gas mixtures under pressure. The process is based on the adsorption principle (adsorption is the binding of particles to the surface of a substance). Special porous materials (zeolites in oxygen plants) are used as molecular sieves to adsorb molecules according to their adsorption forces and/or their kinetic diameter. Pressure swing adsorption uses the physical property that gases adsorb to surfaces to varying degrees. The separation of nitrogen and oxygen is possible due to the kinetic effect. Zeolites are used as adsorbents for the production of oxygen. In this process, the separation of nitrogen and oxygen takes place via the equilibrium effect. The nitrogen is adsorbed on the zeolites that are used.

Installation of Containerised System:

The containerised system must be installed in a suitable location. The ground / plinth must be level and capable of supporting the container load. The location should be selected to provide adequate unrestricted airflow to ensure sufficient in-feed air is available for the compressors. Location close to air ventilation exhaust grilles should be avoided. If locating the containerised system in a car part, the location should be selected to minimise the potential for the ingestion of engine exhaust fumes.

The cooling air from the compressor after coolers and vent gases from the generators are exhausted externally, so the location should be selected to ensure adequate dissipation to the ambient. Sufficient airflow around the container is required to prevent short circuiting of these exhausts into the compressor air intake grilles located in the sides of the container.

Siting of the container in noise sensitive locations should also be avoided, and ease of access for maintenance should also be ensured. The location should be selected to minimise the length of cable and pipe runs when connecting the system to the hospital electrical and medical gas infrastructure.

Inmatec Generators

Compressed air supply

The temperature of the air at the inlet of the IMT POC med oxygen generator must not exceed 40°C. Within the air compressor, an aftercooler removes the mechanical heat of compression from the infeed air to the PSA generator to ensure excessive air temperatures are avoided.

An excessively high inlet temperature reduces the performance of the oxygen generator and causes damage that is not covered by the manufacturer's warranty. An excessively low inlet temperature can cause frost on some components and damage that is not covered by the manufacturer's warranty.

Operational requirements:

The IMT POC med oxygen generator must be installed in a well-ventilated interior. The ambient temperature must be maintained between +5°C and +40°C. Operation of the IMT POC med oxygen generator at temperatures below +5°C or above +40°C will cause damage not covered by the manufacturer's warranty. Automatic ventilation fans installed in the walls of the containers will prevent an excessive build-up of heat within the container. Automatically enabled electric heaters withing the containers will prevent the temperature from dropping too low.

PSA generator alarms:

The generator control system monitors the proper functioning of the generator. Alarms are triggered in the event of malfunctions, i.e. if previously defined limit values are exceeded or not reached. The alarm message then appears on each page of the control panel. All error messages that occur are automatically saved. The alarms can be read out from the data memory at a later time. All alarm messages are displayed in the alarm menu below.

Alarm menu:

Temperature alarm: An alarm is triggered if the temperature exceeds or falls below the limit value.

Alarm inlet pressure: An alarm is triggered if the inlet pressure exceeds or falls below the limit value.

Alarm outlet pressure: An alarm is triggered if the outlet pressure exceeds or falls below the limit value.

Alarm wait time: An alarm that has already been acknowledged is reported again.

Start delay: Suppresses the first alarms when the generator is restarted.

Machine on alarm switch off: Selection Off / On Generator is switched off in case of alarm.

Each container houses two medical oxygen generation systems for the supply of gaseous oxygen at 95% (+/-3%) purity for medical use. Each system can supply up to 16m3/hr at a supply pressure of 7 Bar.

Both systems are Class IIa Medical Devices, each containing two Inmatec Pressure Swing Adsorption (PSA) medical oxygen generators (model POC 8300).

Each 40' container houses assemblies of a number of individually CE marked components which were assembled at Alternative Heat, Banbridge. This is shown in Figure 1, above. (Need to include photo)

The complete assemblies comply with ISO 7396-1, which specifies a requirement to have three sources of medical gas supply and have been built as Configuration A units.

Compliance is achieved by through the following system configuration:

- Primary source of medical oxygen supply is an Inmatec oxygen PSA generator, with an air compressor, cooler, filtration, pressure vessels and process controls.

- Secondary source of supply is another Inmatec oxygen PSA generator, with an air compressor, cooler, filtration, pressure vessels and process controls, identical system to the primary, but which also includes a pressurised reserve system.

- Reserve System comprised of a booster compressor and four high pressure cylinder bundles, which provide a reserve of high pressure oxygen. A pressure reducing regulator is provided on this high pressure reserve, which allows oxygen to be supplied to the hospital system at the required supply pressure without either compressors or PSA generators running.

The skid contains the following components:

(Refer to PDE drawing reference PD7236101 Rev D attached)

Primary System (A)

1. BOGE SLF 40-3 oil lubricated compressor, a BOGE DS 052/2 refrigeration dryer and a 1000 litre, Galvanised, air receiver AR 001, rated for operation up to 10 Bar. The discharge from the compressor passes through a cyclonic oil/water separator, humid air pre-filter, refrigerant dryer, dry air after filter, 225L carbon tower (CT 001) and a dust filter before pressurising the 1000 litre galvanised vertical air receiver. The system has a 10 Bar Maximum working pressure, with over pressure protection provided by a Pressure Relief Valve (PSV 002) set @11Bar, and will normally operate at 7 Bar to supply air to the Oxygen PSA generator.

2. INMATEC GASE Technology GmbH – POC 8300 (MED); 93% O_2 PSA plant with two parallel 500L vessels filled with zeolite, which separates the air into its constituent gases. The temperature and moisture content of the air is checked prior to entering the generator to prevent possible damage to the zeolite media. A control system sequences various valves, which control the length of time each vessel is on-line or is being regenerated. The on-line vessel removes CO_2 and N_2 from the air to provide 93% (+/-3%) purity oxygen for supply to the hospital oxygen delivery system. This is stored in a 1000 litre pressure vessel with gas analysers for monitoring O_2, CO and CO_2 content. The PSA Supply valve is Ball Valve BV 06(A).

3. After BV 06(A) and BV 06(B) there are 2 x 1000L Oxygen Receivers (OV 03 & OV 04), 10 Bar MWP, operating at 6 Bar, with Medical Particulate Filter, Regulator and stainless-steel pipework to the Customer Supply Valve, V 300. There is a Flowmeter (FM 105) on the two receivers

Machine Parts List

Table 3 Machine Parts List

			Compressor Data - A & B					
Section 01	**Tag**		**Description**	**Manufacturer Ref**	**PN**	**DN**	**V**	**WSE**
AC-01	F 01	1	Filter, Suction	Boge SLF 40-3				
AC-02	PI 02	2	Indicator, Filter Delta P	Boge SLF 40-3	10			
AC-03	PS 03	3	Switch, Filter Delta P	Boge SLF 40-3				
AC-04	V 04	4	Valve, Suction Restrictor	Boge SLF 40-3				
AC-05	V 05	5	Valve, Unloader	Boge SLF 40-3				
AC-06	V 051	5.1	Valve, SOV, Suction Restrictor	Boge SLF 40-3				
AC-07	V 06	6	Valve, Double Non-Return, Rotation	Boge SLF 40-3				
AC-08	M 07	7	Motor, 30 kW	Boge SLF 40-3				
AC-09	P 08	8	Compressor, Oil Screw	Boge SLF 40-3				
AC-10	TT 09	9	Transmitter, Air Discharge Temperature	Boge SLF 40-3				
AC-11	R 10	10	Receiver, Air and oil Separator	Boge SLF 40-3				**YES**
AC-12	S 11	11	Separator, Oil, Air Discharge	Boge SLF 40-3				**YES**
AC-13	PSV 12	12	Valve, Pressure Relief	Boge SLF 40-3				**YES**
AC-14	V 13	13	Valve, Quick Start	Boge SLF 40-3				
AC-15	PCV 21	21	Valve, Back Pressure	Boge SLF 40-3				
AC-16	PT 24	24	Transmitter, Pressure, Discharge	Boge SLF 40-3				**YES**
AC-17	AC 22	22	Cooler, Air Discharge, Aftercooler	Boge SLF 40-3				
AC-18	V 25	25	Valve, Air Discharge	Boge SLF 40-3				
AC-19	V 26	16	Valve, Oil, Temperature Control	Boge SLF 40-3				
AC-20	AC 27	17	Cooler, Oil	Boge SLF 40-3				
AC-21	F 28	18	Filter, Oil	Boge SLF 40-3				
AC-22	S 29	29	Separator, Condensate Oil, Refrigerant Air Dryer	Boge SLF 40-3				**YES**

Table 3 Machine Parts List

			Compressor Data - A & B					
	S 61		Cyclone Separator	BOGE	10	50	10	
	F 62		Pre-Filter	BOGE	10	50	10	
	DS 052-2A		Dryer, Refrigeration, R 134a	BOGE	10	50		
	F 63		After Filter	BOGE		50		
	CT 001		Carbon Tower	PCM Scotland, Yarub Turkey	10		255	YES
	F 64		Dust Filter	BOGE		50		
	AR 001		Receiver, Air	BOGE	10		1000	YES
	PSV 002		Receiver, Air, PSV	BOGE	11	12		YES
	PI 003		Indicator, Pressure	BOGE		100		No
			Boge SLF 40-3		7	32		

Table 3 Machine Parts List

Section 02	Tag	Item	Inmatec PSA POC 8300 MED	MED	PN	DN	V	
	A 01	16	Analyser, 02	Ntron SenzTx Zirconia 4 - 20 mA				
	ACO 01	25	Analyser, CO	Inmatec PSA POC 8300 Med				
	ACO2 01	26	Analyser, C02	Inmatec PSA POC 8300 Med				
	ASO 01	15	Analyser, Sensor, O2	Inmatec PSA POC 8300 Med				
	AV 01	9	Vessel, Adsorber	Holzner	11		500	YES
	AV 02	9	Vessel, Adsorber	Holzner	11		500	YES
	CV 01	24	Valve, Check	Inmatec PSA POC 8300 Med				
	DPT 01	**19**	**Analyser, Dew Point, H2O, -80/+50C/ -20/+50C**	**CS Instruments FA 515**				
	F 01	21	Filter, Fine Particles	Inmatec PSA POC 8300 Med				YES
	F 02	22	Filter, Activated Carbon	Inmatec PSA POC 8300 Med				YES
	F 03	14	Filter, Dust	Inmatec PSA POC 8300 Med				
	F 04	27	Filter, Sterile	Inmatec PSA POC 8300 Med				YES
	FC 01	6	Flow Restrictor, Choke Disc	Inmatec PSA POC 8300 Med				
	FC 02	6	Flow Restrictor, Choke Disc	Inmatec PSA POC 8300 Med				
	FC 03	6	Flow Restrictor, Choke Disc	Inmatec PSA POC 8300 Med				
	FC 04	6	Flow Restrictor, Choke Disc	Inmatec PSA POC 8300 Med				
	FM 01	20	Flow Meter	Inmatec PSA POC 8300 Med				
	LOP 01		Local Operation Panel	Inmatec PSA POC 8300 Med				
	PCV 01	1	Valve, Pressure Regulator	Inmatec PSA POC 8300 Med				
	PCV 02	1	Valve, Pressure Regulator	Inmatec PSA POC 8300 Med				
	PCV 03	1	Valve, Pressure Regulator	Inmatec PSA POC 8300 Med				
	PI 01	7	Pressure Indicator	Inmatec PSA POC 8300 Med				
	PI 02	7	Pressure Indicator	Inmatec PSA POC 8300 Med				
	PI 03	7	Pressure Indicator	Inmatec PSA POC 8300 Med				
	PSV 01	**2**	**Valve, Pressure Safety**	**Nuova General Instruments**	**11**	**12**		**YES**
	PSV 02	**2**	**Valve, Pressure Safety**	**Nuova General Instruments**	**11**	**12**		**YES**
	PT 01	**18**	**Transmitter, Pressure**	**Burkert 8316**				

Table 3 Machine Parts List

Section 02	Tag	Item	Inmatec PSA POC 8300 MED	MED	PN	DN	V	
	PV 1	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 2	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 3	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 4	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 5	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 6	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 7	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 8	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 9	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 10	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	PV 11	5	Valve, Pneumatic	Inmatec PSA POC 8300 Med				
	S 01	8	Silencer	Inmatec PSA POC 8300 Med				
	TT 01	17	Transmitter, Temperature	Inmatec PSA POC 8300 Med				
	V 01	**11**	**Vessel, Oxygen Buffer**	**VIG - Magnet**	**10**		**1000**	**YES**
	PSV 02	**2**	**PSV, Oxygen Receiver**	**Nuova General Instruments**	**11**	**25**		**YES**
	NV 01	13	Valve, Adjusting	Inmatec PSA POC 8300 Med				
	NV 02	13	Valve, Adjusting	Inmatec PSA POC 8300 Med				
	NV 03	13	Valve, Adjusting	Inmatec PSA POC 8300 Med				
	BV 01	10	Valve, Ball	Inmatec PSA POC 8300 Med				
	BV 02	10	Valve, Ball	Inmatec PSA POC 8300 Med				
	BV 03	10	Valve, Ball	Inmatec PSA POC 8300 Med				
	BV 04	10	Valve, Ball	Inmatec PSA POC 8300 Med				
	BV 05	10	Valve, Ball	Inmatec PSA POC 8300 Med				
	BV 06	10	Valve, Ball	Inmatec PSA POC 8300 Med				
	SV 01	12	Valve, Pneumatic Bleed	Inmatec PSA POC 8300 Med				
	SV 02	12	Valve, Pneumatic Bleed	Inmatec PSA POC 8300 Med				
	DV01	23	Display, Valve	Inmatec PSA POC 8300 Med				
	PNV 01	3	Valve, Control	Inmatec PSA POC 8300 Med				
	PNV 02	4	Valve, Control	Inmatec PSA POC 8300 Med				

Table 3 Machine Parts List

Section 03	Tag		Oxygen Buffer	Manufacturer Ref	PN	DN	V	WSE
	OV 03		**O2 Vessel 03**	VIG - Magnet	**10**		**1000**	**YES**
	V 100		Valve O2 Vessel 03, Inlet					
	PSV 101		Valve O2 Vessel 03, PSV	**Nuova General Instruments**	**11**	**25**		**YES**
	FM 105	**VA 520**	**Flow Meter**	**CS Instruments**		12		
	OV 04		**O2 Vessel 04**	VIG - Magnet	**10**		**1000**	**YES**
	PSV 111		Valve O2 Vessel 04, PSV	**Nuova General Instruments**	**11**	**25**		**YES**
	NRV 121		Valve, Non Return					
Sauerstoff	**PCV 112**	**DRV 2****	**Valve, Pressure Regulator**	**Landefeld**		**12**		
	PT 123		Transmitter, Pressure, Supply					

Table 3 Machine Parts List

Section 04	Tag		Reserve Oxygen System	Manufacturer Ref	PN	DN	V	WSE
	V 200		Valve, Booster Inlet		10			
	F 201		Filter, Booster Inlet		10			
	PCV 202		Valve, Back Pressure		10			YES
	C 209		Compressor, 3 Stage, 4.15 kW, 200Bar	OxySystems	200			YES
	PSL 107		Switch Pressure, LO, Start		200			
	PSH 208		Switch Pressure, HI, Stop		200			
	PT 206		Transmitter, Pressure, Reserve Buffer		200			YES
	V 204		Valve, Isolation, High Pressure		200			
	NRV 205		Valve, Non Return HP		200			
	PSV 203		Valve, Pressure Relief		200			YES
Sauerstoff	**PCV 210**	**DR 14-7 HD**	**Valve, Pressure Regulator**	**Messing - Brass**	**200**			
	PSV 211		Valve Pressure Relief		11			YES
	SV 212		Valve, Solenoid Operated, Reserve		10			
	V 213		Valve, Manual, Isolation		10			
	V 214		Valve, Manual, Vent		10			
	HC 221		Pigtail Assembly, Buffer		200			
	HC 222		Pigtail Assembly, Buffer		200			
	HC 223		Pigtail Assembly, Buffer		200			
	HC 224		Pigtail Assembly, Buffer		200			
	HP B01		TPED Bundle 12 x 50L WC, 200 Bar MWP	Wystrach	200		600	YES
	HP B02		TPED Bundle 12 x 50L WC, 200 Bar MWP	Wystrach	200		600	YES
	HP B03		TPED Bundle 12 x 50L WC, 200 Bar MWP	Wystrach	200		600	YES
	HP B04		TPED Bundle 12 x 50L WC, 200 Bar MWP	Wystrach	200		600	YES
	V 300		**Valve, Med O2 Supply**		**10**			

Section 4 Risk Assessment & ESR

Table 4 ISO 12100 2010 Machine Safety - Risk Assessment

	Stage 1	*HAZARD IDENTIFICATION*	*ISO 12100: 2010*					
	20/04/2020	*Peter Watters CEng FIMechE FIEI*	*PCM Risk Assessment*		*L, M, H*	*L, M, H*	*1 - 9*	
Type	*Hazards*	*Origin*	*Potential Consequence*	*Risks*	*Severity*	*Likelihood*	*RISK*	
1	Mechanical	Rough, slippery Surface	Slipping, Tripping and Falling	Confined Space in Container	M	M	4	ESR 01
2	Electrical	Live parts	Electrocution	400 V AC Control Panel	H	L	3	ESR 02
2	Electrical	Parts live under Fault	Fire	Electrical Fire in Container	M	L	2	ESR 03
3	Thermal	Objects or materials with HI/LO temperature	Burns	Compressor Surfaces > 40C	L	M	2	ESR 04
4	Noise Hazards	Moving Parts	Discomfort	2 Compressor in Container, PPE	L	M	2	ESR 05
7	Material/Substance	Biological & microbiological agent	Breathing difficulties	Cleanliness, Oil Filter, Filters	M	L	2	ESR 06
7	Material/Substance	Gas	Explosion	155 Bar Oxygen System	L	M	2	ESR 07
7	Material/Substance	Oxidizer	Fire	Oxygen Enriched Air,	L	M	2	ESR 08
8	Ergonomic	Access	Musculoskeletal disorder	Shipping Container, 1 Access Door	L	M	2	ESR 09
8	Ergonomic	Local Lighting	Stress	Container & Emergency Lights	L	M	2	ESR 10
9	Environment	Pollution	Any other as a consequence of the effect caused by the sources of hazard of the machine	Air Intakes to Plant and Nitrogen Vents	L	L	1	ESR 11
10	Combination	e.g High Temp & Effort	Eg dehydration, heat stroke	Working in Shipping Container	L	M	2	ESR 12

Section 5 ESR Residual Risks

The Machinery Risk Assessment identified twelve residual risks to be covered in the plant operating instructions.

Table 5 Residual Risks from the Plant Design

No	Hazards	Risks	Management
1	Rough, slippery Surface	Confined Space in Container	Container Design, lighting ventilation and PPM tasks
2	Live parts	400 V AC Control Panel	Panel design and test
3	Parts live under Fault	Electrical Fire in Container	Panel design and test, cooling fans, alarms and fire suppression
4	Objects or materials with HI/LO temperature	Compressor Surfaces > 40C	Container layout around HP compressor & Cooling
5	Moving Parts	2 Compressor in Container, PPE	
6	Biological & microbiological agent	Cleanliness, Oil Filter, Filters	System cleanliness in installatio, test and commissioning.
7	Gas	155 Bar Oxygen System	HP plant design and test
8	Oxidizer	Oxygen Enriched Air,	Gas analysis for leaks. Intake and vent design
9	Access	Shipping Container, 1 Access Door	One access door, with Container open for maintenance. Building Regs Plant room,
10	Local Lighting	Container & Emergency Lights	Container Design and layout
11	Pollution	Air Intakes to Plant and Nitrogen Vents	Container in planned location, away from traffic and vehicle exhausts
12	Rough, slippery Surface	Working in Shipping Container	PPM requirement and housekeeping in container.

Written Scheme of Examination (WSE)

Each Container is a Pressure system and the Planned Preventative Maintenance will be based on the Written Scheme of Examination for this Intermediate category Pressure System. Table 3 has identified the items subject to WSE

Section 6 Operating Instructions

1. Container

The container is a mobile plant room, with entry by a door at one end. Inside there is a Smoke/Fire detection system, Emergency lights, heaters and cooling fans.

The Container doors must be open for any work to be done inside

2. 400V AC Power Supply

The container has a 400V AC control Panel on the left-hand side of the entry door. This has a mains isolator and there are indicators and controls on the panel door.

3. Compressor Operation

The Compressor and Refrigeration dryer are designed to run automatically to maintain dry air in the air receiver. The compressor data is as follows.

Table 6.1 – Compressor Operation

Control	Function	Set Point	Remark
Local Operator Panel	Display & Controls		Ethernet & Fieldbus
HI Pressure	Compressor Stops		Target Pressure, No Flow
HIHI Pressure	Compressor Trips		Compressor Trip
LO Pressure	Low Pressure Alarm		System Fault
Dew Point	Target Range		Water Content
HI Delta P	Filter Monitoring		Filter Exchange
HIHI Temperature	Compressor		Compressor Trip
Primary/Secondary	Duty Selector		Change-over

NB: The compressor System is Validated for the Hospital Pharmacist to Show the set points are achieved and maintained. In a fault condition the back-up compressor will start automatically.

4. Oxygen Generator

Table 6.2 – PSA Operation

Control	Function	Set Point	Remark
Inlet Pressure			
Dew Point	Target Range		Water Content
Inlet Temperature	Min. Temperature		System Fault
Dew Point	Target Range		Water Content
Flow	Discharge Flow rate		
Local Operator Panel	Display & Controls		Ethernet & Fieldbus
Outlet Purity	Oxygen		
Outlet Purity	CO		
Outlet Purity	CO2		
Outlet Pressure			PT 123

NB: The Oxygen PSA System has been approved by NB 1250 as a Medical Device and is validated for the Hospital Pharmacist to show the set points are achieved and maintained. In a fault condition the back-up compressor will start automatically.

5. Reserve Oxygen System

Table 6.3 – Reserve System Operation

Control	Function	Set Point	Remark
V 200	Inlet Valve		
PSL 207	LO Pressure		Compressor Start
PSH 208	HI Pressure		Compressor Stop
PT 206	System Pressure	50% of 155Bar	Low Pressure Alarm
PCV 210	HI Pressure Regulator	155Bar/7Bar	HP supply regulator
PSV 211	Low Pressure Safety Valve	7 Bar	Low Pressure Protection
SV 212	Solenoid Valve		Ethernet & Fieldbus

NB: The High-Pressure compressor system is validated for the hospital pharmacist to show the set points are achieved and maintained. The operation of the High-Pressure system is also validated.

6. Emergency Instructions

- The two BOGE Compressors and the HP Compressor have individual EMERGENCY STOPS to stop the energy source to the pressure system. These can be reset, individually on each machine.
- The Emergency Stop will show on the controls as Fault.
- SV 212 does not have an emergency stop function.
- Valves BV 06A & BV 06B and V 213 are the part System Isolation Valves.
- Valve V 300 Isolates the Machine.

7. Planned Preventative Maintenance and WSE

The Planned Preventative Maintenance Schedule and WSE are separate document based on Table 3.

Section 7 EC Declaration of Conformity

Declaration of Conformity

CE Marking

- PCM Engineering Services (Belfast) Ltd. Unit 3, 38 Duncrue Crescent, Belfast BT3 9BW
- CE Mark, 2020
- Medical Oxygen Supply Skid
- Serial number 2020_01 & _02

Dr Peter Watters CEng FIMechE FIEI, authorised representative of the Manufacturer, PCM Air Power, has Certified Conformity of Machine by the assessment of conformity, by internal production control, during the manufacture and as Pressure Equipment Directive responsible person.

The following Directives were followed to complete the internal production control process described in PED Sound Engineering Practice and in the Machinery Directive, to conform to the following directives.

- 2014/68 EC Pressure Equipment Directive (PED)
- 2014/30/EC Electromagnetic Compatibility Directive (EMC)
- 2006/42/EC Machinery Directive (MD)
- 2014/35/EC Low Voltage Directive (LVD

Date: 10 April 2020

Signed:

PCM Engineering Services (Belfast) Ltd.

Unit 3, 38 Duncrue Crescent

Belfast

BT3 9BW

Section 8 References and Documents

MSDS Sheets
Provided separately for compressor oils, activated carbon, oxygen and zeolite

EU “New Approach” Directives

These EU directives are linked to a list of EN and other standards on the EU New Approach Web site. The machine design was built from the annexes of each of the four directives, and these standards are references.

http://www.newapproach.org/Directives/DirectiveList.asp

Supporting Documents

Name	Description	Rev	Date
R(EU) 2017/745	Regulation on Medical Devices		05 Apr 2017
ISO 7396_1	Medical Gas Pipeline Systems; Pipeline systems for compressed medical gases and vacuum		
ISO 14971	Medical Devices; Application of Risk Management to medical devices.		
	BOGE Compressor Manual		
	BOGE Dryer Manual		
E1719303	InMatec On Touch-POC Med Drawing	Pro	24 Feb 2020
	InMatec Operating Manual	004.2019	19 Mar 2019
	Green Electrical Drawing		